

Cambridge International AS & A Level Mathematics 9709

Pure Mathematics 3 | P3 九单元六段式双语系统教材

适用考纲 | Syllabus: 2028–2030 (核心内容兼容 2026–2027) | Edition: 2026.09

Paper 3: 1 hour 50 minutes, 75 marks, 9–11 structured questions. P1 knowledge is assumed.
本书题目与数字均为原创 Cambridge-style 训练, 不复制真题原文。

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3.1 | 代数 | Algebra

1. 单元导读 | Unit introduction

本单元严格对应官方 3.1。学习时先建立概念与条件，再推导公式，随后通过完整 worked examples 把方法转化为可复现步骤。P3 的得分关键是：选择正确结构、展示 decisive working、保持 exact values，并检查定义域、区间、参数或几何意义。

与国内课程衔接 | Connection. 国内高中教材已提供相近的代数与微积分基础；Cambridge 更强调多步骤结构题、英文命令词、前后小问的 hence 连接，以及对限制条件的显式说明。

2. 知识点精讲 | Detailed teaching (共 7 个知识点)

知识点 01/07 | 绝对值是距离，不只是“去掉负号” | Modulus as distance

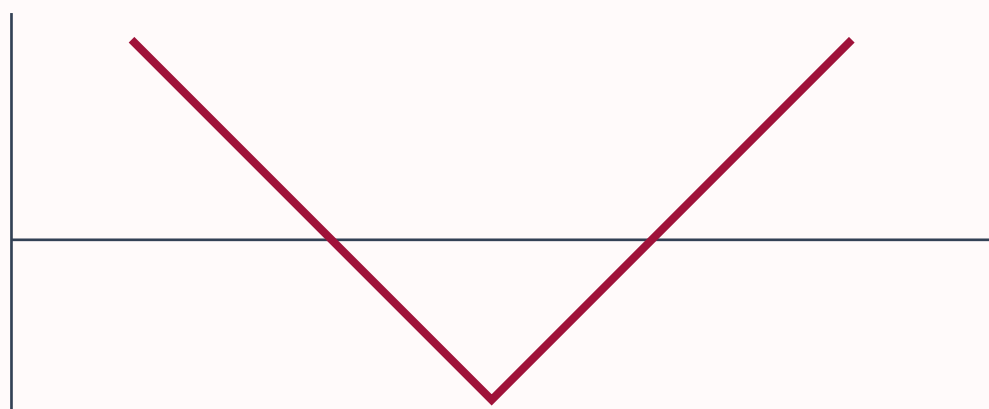
$|x-a|$ 表示 x 到 a 的距离，所以 $|x-a|<b$ ($b>0$) 等价于 $a-b<x<a+b$ ；若 $b\leq 0$ 必须先判断是否可能。方程 $|u|=|v|$ 可平方为 $u=v$ ，等价于 $u=v$ 或 $u=-v$ 。图像 $y=|ax+b|$ 在 $ax+b=0$ 处折返，顶点为 $(-b/a, 0)$ 。

English explanation and method. Worked example. Solve $|3x-2|=|x+4|$. Squaring gives $(3x-2)^2=(x+4)^2$, so $3x-2=x+4$ or $3x-2=-(x+4)$. Hence $x=3$ or $x=-1/2$. Both satisfy the original equation. For an inequality with an outside term, first isolate the modulus and use sign regions; do not apply a memorised interval rule blindly.

知识点 02/07 | 多项式除法与恒等式检查 | Polynomial division

若 $P(x)$ 除以 $D(x)$ ，则 $P(x)=D(x)Q(x)+R(x)$ ，且 $\text{degree } R < \text{degree } D$ 。长除法每一步都消去当前最高次项；若某次幂缺项，必须写 $0x$ 占位。

English explanation and method. Worked example. Divide $2x^2-3x+4$ by $x-2$. Synthetic or long division gives quotient $2x+x+2$ and remainder 8 because $(x-2)(2x+x+2)+8=2x^2-3x+4$. The final identity is the fastest error check.



$y = |ax+b|$; vertex where $ax+b=0$

知识点 03/07 | 余式定理、因式定理与非首一因式 | Remainder and factor theorems

$P(x)$ 除以 $x-a$ 的余数是 $P(a)$ 。若因式为 $ax+b$ ，零点是 $x=-b/a$ ；因此检验 $2x-3$ 是否为因式要算 $P(3/2)$ ，不是 $P(3)$ 。 $P(a)=0$ 当且仅当 $x-a$ 是因式。

English explanation and method. Example. If $P(x)=2x^2+kx-5x+6$ has factor $2x-3$, substitute $x=3/2$: $2(27/8)+k(9/4)-15/2+6=0$. This simplifies to $21/4+9k/4=0$, so $k=-7/3$. Show the substitution: the value of k is not a calculator-only result.

知识点 04/07 | 部分分式先由分母结构决定形式 | Choosing a partial-fraction form

三个不同线性因式用 $A/(ax+b)+B/(cx+d)+C/(ex+f)$; 重因式 $(cx+d)$ 必须同时出现 $B/(cx+d)+C/(cx+d)$; 不可约二次因式 $cx+d$ 的分子必须是 $Bx+C$ 。题目范围内分子次数低于分母次数。

English explanation and method. Worked example. $(5x+1)/[(x-1)(x+2)] = A/(x-1)+B/(x+2)$. Multiplying through gives $5x+1=A(x+2)+B(x-1)$. Put $x=1$: $A=2$. Put $x=-2$: $B=3$. Thus the decomposition is $2/(x-1)+3/(x+2)$. Substituting a neutral value such as $x=0$ checks the result.

知识点 05/07 | 有理指数二项式与收敛区间 | Rational binomial expansion

对 rational n , $(1+u)^n = 1+nu+n(n-1)u^2/2!+n(n-1)(n-2)u^3/3!+\dots$, 条件是 $|u|<1$ 。若原式不是 $1+u$, 先提出常数; 有效区间必须从 $|u|<1$ 反解。

English explanation and method. Worked example. Expand $(2-x)^{-1}$ to x . Write $(2-x)^{-1} = 1/2(1-x/2)^{-1}$. Hence $=1/2[1+x/2+x^2/4+\dots] = 1/2+x/4+x^2/8+\dots$. Validity is $|x/2|<1$, so $|x|<2$. The outside factor $1/2$ and the interval are both examinable.

知识点 06/07 | 含绝对值不等式的分区法 | Sign-region method

若不等式两边都含 x , 最稳妥的方法是找出每个绝对值内部为零的临界点, 将数轴分区。每一区间内按实际符号去掉绝对值, 解线性不等式后再与该区间取交集, 最后合并。

English explanation and method. Worked example. Solve $|x-1|>x+3$. When $x+3<0$, i.e. $x<-3$, the inequality is automatically true because the left side is non-negative. For $x\geq-3$, both sides may be squared: $(x-1)^2>(x+3)^2$, giving $-8x>8$ and $x<-1$. Combining with $x\geq-3$ gives $[-3,-1)$, together with $x<-3$, hence $x<-1$.

知识点 07/07 | 代数答案的三重核验 | Three checks for algebra

部分分式用重新合并检查; 多项式除法用 $P=DQ+R$ 检查; 二项式用 $x=0$ 的常数项和一次项符号检查。三个检查分别针对漏项、余数错误和提出常数错误。

English explanation and method. A high-quality P3 solution makes the algebra reversible. If the decomposition is correct, recombining it reproduces the numerator identically. If the expansion is correct, its validity interval is stated separately from the approximation itself.

3. 核心词汇表 | Core vocabulary (共 10 词)

English term	中文准确口径	Exam-ready meaning
modulus	绝对值	the distance of a real number from zero.
polynomial	多项式	an expression in non-negative integer powers of a variable.
quotient	商	the result after division excluding the remainder.
remainder theorem	余式定理	the remainder on division by $x-a$ is $P(a)$.
factor theorem	因式定理	$x-a$ is a factor exactly when $P(a)=0$.
partial fraction	部分分式	a simpler fraction used in a decomposition.
repeated factor	重因式	a factor occurring more than once.
irreducible quadratic	不可约二次因式	a real quadratic with no real linear factors.
binomial expansion	二项式展开	a power series expansion of a two-term expression.
validity interval	有效区间	the values for which an infinite expansion is valid.

4. 单元练习 | Unit practice (共 8 题)

练习 01/08

Question. Solve $|2x-5|<7$. [3]

练习 02/08

Question. Divide $x+2x-5x-6$ by $x+3$. [4]

练习 03/08

Question. $P(x)=2x+kx-5x+6$ has factor $2x-3$. Find k . [4]

练习 04/08

Question. Express $(7x+5)/[(x+1)(x+2)]$ in partial fractions. [4]

练习 05/08

Question. Find the first three terms of $(1-2x)^{-1/2}$ and state validity. [5]

练习 06/08

Question. Explain why squaring a modulus equation is safe but squaring a general equation may introduce roots. [3]

练习 07/08

Question. Solve $|x-1|>x+3$. [5]

练习 08/08

Question. Find the remainder when $3x-2x+5$ is divided by $2x+1$. [3]

5. 练习答案与解析 | Answers and explanations (共 8 题)

答案 01/08

Correct answer and working. $-7<2x-5<7$, so $-2<2x<12$ and therefore $-1<x<6$.

答案 02/08

Correct answer and working. Long division gives $x-x-2$ with remainder 0. Check:
 $(x+3)(x-x-2)=x+2x-5x-6$.

答案 03/08

Correct answer and working. $P(3/2)=0$ gives $27/4+9k/4-15/2+6=0$, hence $21+9k=0$ and $k=-7/3$.

答案 04/08

Correct answer and working. Let $A/(x+1)+B/(x+2)$. Then $7x+5=A(x+2)+B(x+1)$. $x=-1$ gives $A=-2$; $x=-2$ gives $B=9$.
Answer: $-2/(x+1)+9/(x+2)$.

答案 05/08

Correct answer and working. Use $n=-1/2, u=-2x$: $1+(-1/2)(-2x)+[(-1/2)(-3/2)/2](4x)=1+x+3x/2+\dots$. Valid when $|-2x|<1$, so $|x|<1/2$.

答案 06/08

Correct answer and working. $|u|=|v|$ is exactly equivalent to $u=v$ because both sides are non-negative. In a general equation $u=v$, squaring loses sign information and also permits $u=-v$, so candidate roots must be checked.

答案 07/08

Correct answer and working. For $x < -3$ the right side is negative, so all such x work. For $x \geq -3$, squaring gives $(x-1) > (x+3)$, hence $x < -1$. Combining gives $x < -1$.

答案 08/08

Correct answer and working. The zero of $2x+1$ is $-1/2$. $R=P(-1/2)=3/16+1+5=99/16$.

6. 自学检查清单 | Self-study checklist (共 10 项)

- ☐ 能准确说明 绝对值是距离，不只是“去掉负号”
- ☐ 能准确说明 多项式除法与恒等式检查
- ☐ 能准确说明 余式定理、因式定理与非首一因式
- ☐ 能准确说明 部分分式先由分母结构决定形式
- ☐ 能准确说明 有理指数二项式与收敛区间
- ☐ 能准确说明 含绝对值不等式的分区法
- ☐ 能准确说明 代数答案的三重核验
- ☐ 能在没有答案提示时完成本单元全部练习并保留关键步骤
- ☐ 能检查定义域、区间、符号、精度或几何意义，而不是只抄计算器数值
- ☐ 能用英文说明所选公式或方法为何适用于当前条件

3.2 | 对数与指数函数 | Logarithmic and exponential functions

1. 单元导读 | Unit introduction

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2. 知识点精讲 | Detailed teaching (共 7 个知识点)

知识点 01/07 | 指数与对数互为逆运算 | Indices and logarithms

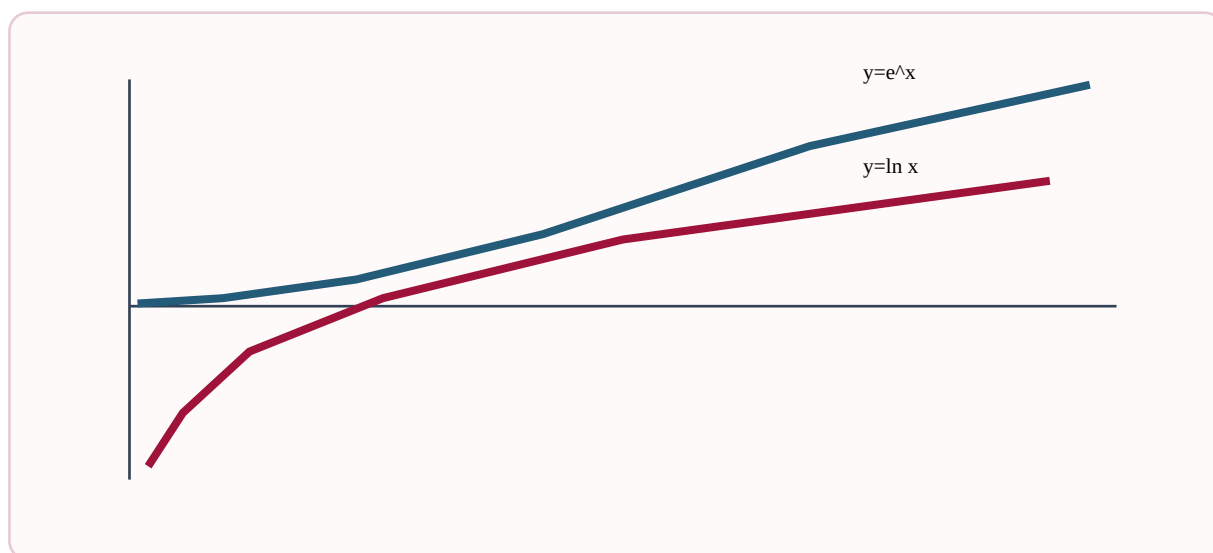
$a=y$ 等价于 $\log y=x$ ；在 P3 中重点使用 e 与 $\ln x$ 。 $\ln x$ 只在 $x>0$ 定义， e 对所有实数有定义且恒为正； $\ln(e)=x$ ，而 $e^{(\ln x)}=x$ 需要 $x>0$ 。

English explanation and method. The graph $y=e^x$ passes through $(0,1)$ and has horizontal asymptote $y=0$. The graph $y=\ln x$ passes through $(1,0)$ and has vertical asymptote $x=0$. They are reflections in $y=x$ because they are inverse functions.

知识点 02/07 | 对数定律必须保留定义域 | Log laws and domains

$\ln(ab)=\ln a+\ln b$ ， $\ln(a/b)=\ln a-\ln b$ ， $\ln(a)=k \ln a$ ，但所有被取对数的量必须为正。 $\ln(a+b)$ 不能拆开。解方程前后都应检查 log arguments。

English explanation and method. Worked example. Solve $\ln(x-1)+\ln(x+2)=\ln 12$. Domain: $x>1$. Combine to $\ln[(x-1)(x+2)]=\ln 12$, so $x^2+x-14=0$. The roots are $(-1 \pm \sqrt{57})/2$; only $(-1+\sqrt{57})/2$ is greater than 1, so the other root is rejected.



知识点 03/07 | 未知数在指数中的方程与不等式 | Exponential equations

把两边写成同底数，或在正值两边取 $\ln$ 。处理不等式时，底数 $e>1$ ，所以 $\ln$ 保持方向；若用 $0<a<1$ 的指数函数比较，函数递减会改变方向。

English explanation and method. Example. $3^{(2x-1)}=7$ gives $(2x-1)\ln 3=\ln 7$, so $x=(1+\ln 7/\ln 3)/2$. Keep the exact logarithmic form unless a decimal accuracy is requested.

知识点 04/07 | 幂律关系的线性化 | Linearising $y=kx$

若 $y=kx$ ($k>0, x>0$)，取 $\ln$ 得 $\ln y=\ln k+n \ln x$ 。以 $X=\ln x$ 、 $Y=\ln y$ 作图， $\text{gradient}=n$ ， $\text{vertical intercept}=\ln k$ ，因此 $k=e^{(\text{intercept})}$ 。

English explanation and method. Worked example. A $\ln y$ against $\ln x$ graph has gradient 1.6 and intercept 0.7. Then $n=1.6$ and $\ln k=0.7$, so $k=e^{0.7}$. The model is $y=e^{0.7} x^{1.6}$, not $y=0.7x^{1.6}$.

知识点 05/07 | 指数关系的线性化 | Linearising $y=ka$

$y=ka$ 取 $\ln$ 得 $\ln y=\ln k+x \ln a$ 。画 $\ln y$ 对 x ，斜率是 $\ln a$ ，截距是 $\ln k$ ；故 $a=e^{(\text{gradient})}$ 、 $k=e^{(\text{intercept})}$ 。

English explanation and method. If the gradient is negative, then $0<a<1$ and the model represents exponential decay. Always state which transformed variables are plotted: a straight line on ordinary y against x axes does not follow from this transformation.

知识点 06/07 | 指数模型中的倍增与半衰期 | Doubling and half-life

若 $y=Ae^{(kt)}$ ，倍增时间 T 满足 $e^{(kT)}=2$ ，因此 $T=\ln 2/k$ ($k>0$)；若衰减 $y=Ae^{(-kt)}$ ，半衰期 $T=\ln 2/k$ 。A 只改变初始量，不改变倍增或半衰时间。

English explanation and method. Example. A culture follows $N=240e^{(0.18t)}$. The doubling time is $\ln 2/0.18 \approx 3.85$ time units. This result comes from $480=240e^{(0.18T)}$; cancelling the initial amount explains why the doubling time is independent of 240.

知识点 07/07 | 对数不等式与单调性 | Logarithmic inequalities

$\ln x$ 在 $x>0$ 上严格递增，所以对正的 A 、 B ， $\ln A<\ln B$ 等价于 $A<B$ 。指数函数 e 也递增；但变形过程中必须先保留每个对数的正值条件。

English explanation and method. Example. $\ln(2x-1)<\ln 5$ requires $x>1/2$ and then $2x-1<5$, giving $1/2<x<3$. Writing only $x<3$ ignores the logarithm domain.

3. 核心词汇表 | Core vocabulary (共 10 词)

English term	中文准确口径	Exam-ready meaning
index	指数	the power to which a base is raised.
logarithm	对数	the inverse operation of exponentiation.
natural logarithm	自然对数	logarithm to base e .
exponential function	指数函数	a function with the variable in the exponent.
inverse function	反函数	a function reversing another function.
asymptote	渐近线	a line approached by a curve.
log law	对数定律	a rule converting products, quotients or powers.
linearisation	线性化	transforming a relationship into straight-line form.
gradient	斜率	change in vertical variable per unit horizontal change.
intercept	截距	the value where a graph crosses an axis.

4. 单元练习 | Unit practice (共 8 题)

练习 01/08

Question. Solve $\ln(x-1)+\ln(x+2)=\ln 12$. [5]

练习 02/08

Question. Solve $5^{2x-1}=11$, giving an exact answer. [3]

练习 03/08

Question. A graph of $\ln y$ against $\ln x$ has gradient 2.4 and intercept -0.3 . Find y in terms of x . [4]

练习 04/08

Question. A graph of $\ln y$ against x has gradient -0.18 and intercept 1.6. Express $y=ka^x$. [4]

练习 05/08

Question. State the domain and range of $y=\ln(3-x)$. [3]

练习 06/08

Question. Explain the error in $\ln(x+4)=\ln x+\ln 4$. [2]

练习 07/08

Question. Solve $\ln(2x-1)<\ln 5$. [3]

练习 08/08

Question. $N=240e^{(0.18t)}$. Find the doubling time. [3]

5. 练习答案与解析 | Answers and explanations (共 8 题)

答案 01/08

Correct answer and working. Domain $x>1$. Combine logs: $(x-1)(x+2)=12$, so $x^2+x-14=0$. Hence $x=(-1 \pm \sqrt{57})/2$; only $x=(-1+\sqrt{57})/2$ satisfies $x>1$.

答案 02/08

Correct answer and working. Taking $\ln$: $(2x-1)\ln 5=\ln 11$, so $x=[1+\ln 11/\ln 5]/2$.

答案 03/08

Correct answer and working. $\ln y=2.4\ln x-0.3$. Thus $n=2.4$ and $k=e^{(-0.3)}$, giving $y=e^{(-0.3)}x^{2.4}$.

答案 04/08

Correct answer and working. $\ln y=1.6-0.18x$, so $k=e^{1.6}$ and $\ln a=-0.18$, hence $a=e^{(-0.18)}$. Therefore $y=e^{1.6}(e^{-0.18})^x$.

答案 05/08

Correct answer and working. Need $3-x>0$, so domain $x<3$. The natural logarithm can take every real value, so range is all real numbers.

答案 06/08

Correct answer and working. The product law applies to $\ln(4x)$, not a sum inside a logarithm. There is no law splitting $\ln(x+4)$.

答案 07/08

Correct answer and working. Domain $x > 1/2$. Since $\ln$ is increasing, $2x-1 < 5$ gives $x < 3$. Hence $1/2 < x < 3$.

答案 08/08

Correct answer and working. Set $480 = 240e^{(0.18T)}$. Then $e^{(0.18T)} = 2$, so $T = \ln 2 / 0.18 \approx 3.85$.

6. 自学检查清单 | Self-study checklist (共 10 项)

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- ☐ 能准确说明 对数定律必须保留定义域
- ☐ 能准确说明 未知数在指数中的方程与不等式
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3.3 | 三角函数 | Trigonometry

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2. 知识点精讲 | Detailed teaching (共 7 个知识点)

知识点 01/07 | 六个三角函数与定义域 | Six trigonometric functions

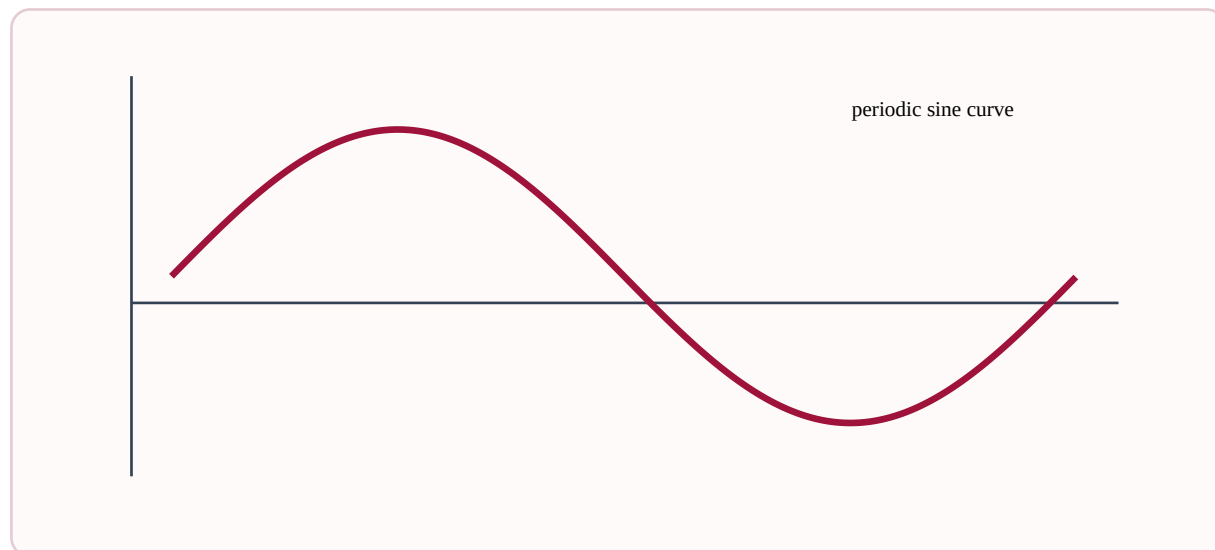
$\sec x = 1/\cos x$, $\operatorname{cosec} x = 1/\sin x$, $\cot x = \cos x/\sin x = 1/\tan x$ 。零点、周期和垂直渐近线从倒数关系得到：例如 $\sec x$ 在 $\cos x = 0$ 即 $x = \pi/2 + k\pi$ 处无定义。

English explanation and method. Sketches must show period, intercepts and asymptotes. Reciprocal does not mean inverse function: $\sec x$ is $1/\cos x$, whereas $\cos x$ denotes a principal inverse relation.

知识点 02/07 | 恒等式的选择来自目标形式 | Selecting identities

基础恒等式为 $\sin x + \cos x = 1$ 、 $1 + \tan x = \sec x$ 、 $1 + \cot x = \operatorname{cosec} x$ 。证明恒等式时通常只变一边；含 $\sec/\tan$ 时可转成 $\sin/\cos$ ，或使用配套平方恒等式。

English explanation and method. Example. $(\sec x - \tan x)(\sec x + \tan x) = \sec^2 x - \tan^2 x = 1$. State restrictions if division is involved; an algebraic identity is valid only where both original sides are defined.



知识点 03/07 | 和差角与二倍角 | Compound and double angles

$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$; $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$; $\tan(A \pm B) = (\tan A \pm \tan B) / (1 \mp \tan A \tan B)$ 。由 $A=B$ 得二倍角，尤其 $\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$ 。

English explanation and method. Worked example.

$\cos 75^\circ = \cos(45^\circ + 30^\circ) = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} - \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = (\sqrt{6} - \sqrt{2})/4$. Exact-value questions require radicals, not a premature decimal.

知识点 04/07 | $R\sin(x + \alpha)$ 或 $R\cos(x - \alpha)$ | Harmonic form

$a \sin x + b \cos x = R \sin(x + \alpha) = R \sin x \cos \alpha + R \cos x \sin \alpha$, 所以 $R \cos \alpha = a$ 、 $R \sin \alpha = b$, $R = \sqrt{a^2 + b^2}$, 并用 a 、 b 的符号确定 α 的象限。

English explanation and method. Example. $3 \sin x + 4 \cos x = 5 \sin(x + \alpha)$, where $\cos \alpha = 3/5$ and $\sin \alpha = 4/5$. Its maximum is 5 and minimum is -5. To solve $3 \sin x + 4 \cos x = 2$, solve $\sin(x + \alpha) = 2/5$ and then list every solution in the stated interval.

知识点 05/07 | 三角方程必须列全区间解 | Solving trig equations

先化为单一函数, 再求 reference angle, 按象限与周期列出全部解。若平方或除以 $\sin x / \cos x$, 可能引入根或丢失使除数为零的根。

English explanation and method. Worked example. Solve $2 \cos x - 3 \sin x = 0$ for $0 \leq x \leq 2\pi$. Replace $\cos x$ by $1 - \sin x$: $2 \sin x + 3 \sin x - 2 = 0 = (2 \sin x - 1)(\sin x + 2)$. Thus $\sin x = 1/2$, giving $x = \pi/6, 5\pi/6$; $\sin x = -2$ is impossible.

知识点 06/07 | 用辅助角求范围与方程根 | Range and equations using R form

将 $a \sin x + b \cos x$ 写成 $R \sin(x + \alpha)$ 后, 范围立即是 $[-R, R]$ 。若右端常数绝对值大于 R , 则无实根; 若在范围内, 先解移位后的角, 再减 α 并筛选原区间。

English explanation and method. Example. $3 \sin x + 4 \cos x = 6$ has no real solution because the left side has maximum $\sqrt{3^2 + 4^2} = 5$. This one-line range argument is stronger than attempting inverse sine of $6/5$.

知识点 07/07 | 解方程时避免除掉可能为零的因子 | Avoiding lost roots

从 $\sin x \cdot F(x) = 0$ 开始, 若直接除以 $\sin x$, 会丢失 $\sin x = 0$ 的全部根。应先因式分解并分别令每个因子为零。平方后则相反: 可能增加不满足原式的根, 必须回代。

English explanation and method. When transforming a trigonometric equation, record whether the step is reversible. Factoring is reversible; dividing by an expression that may be zero is not. This is a common reason why otherwise accurate solutions miss endpoints or whole families of roots.

3. 核心词汇表 | Core vocabulary (共 10 词)

English term	中文准确口径	Exam-ready meaning
secant	正割	the reciprocal of cosine.
cosecant	余割	the reciprocal of sine.
cotangent	余切	the reciprocal of tangent.
identity	恒等式	an equality true throughout its common domain.
compound angle	和差角	an angle written as a sum or difference.
double-angle formula	二倍角公式	an identity involving twice an angle.
harmonic form	辅助角形式	$R \sin$ or $\cosine$ form for $a \sin x + b \cos x$.
reference angle	参考角	the acute angle used to locate trig solutions.
period	周期	the smallest positive repeat length.
principal value	主值	the selected output of an inverse relation.

4. 单元练习 | Unit practice (共 8 题)

练习 01/08

Question. Prove $(\sec x - \tan x)(\sec x + \tan x) = 1$. [3]

练习 02/08

Question. Find the exact value of $\sin 75^\circ$. [3]

练习 03/08

Question. Write $5\sin x - 12\cos x$ as $R\sin(x - \alpha)$, with α acute. [4]

练习 04/08

Question. Solve $2\cos x - 3\sin x = 0$ for $0 \leq x \leq 2\pi$. [5]

练习 05/08

Question. Solve $\tan x + \cot x = 4$ for $0 < x < \pi/2$. [5]

练习 06/08

Question. State the period and vertical asymptotes of $y = \operatorname{cosec}(2x)$. [3]

练习 07/08

Question. Determine the range of $8\sin x + 15\cos x$. [3]

练习 08/08

Question. Explain why $3\sin x + 4\cos x = 6$ has no real solution. [2]

5. 练习答案与解析 | Answers and explanations (共 8 题)

答案 01/08

Correct answer and working. Difference of squares gives $\sec x - \tan x$. Since $\sec x = 1 + \tan x$, the expression equals 1 wherever the original expressions are defined.

答案 02/08

Correct answer and working. $\sin(45^\circ + 30^\circ) = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ = (\sqrt{6} + \sqrt{2})/4$.

答案 03/08

Correct answer and working. $R=13$. Comparing $13\sin(x - \alpha) = 13\sin x \cos \alpha - 13\cos x \sin \alpha$ gives $\cos \alpha = 5/13$, $\sin \alpha = 12/13$; $\alpha = \tan^{-1}(12/5)$.

答案 04/08

Correct answer and working. Use $\cos x = 1 - \sin x$: $2\sin x + 3\sin x - 2 = 0$, so $(2\sin x - 1)(\sin x + 2) = 0$. Thus $\sin x = 1/2$ and $x = \pi/6, 5\pi/6$.

答案 05/08

Correct answer and working. Let $t = \tan x > 0$. Then $t + 1/t = 4$, so $t^2 - 4t + 1 = 0$ and $t = 2 \pm \sqrt{3}$. Hence $x = \tan^{-1}(2 - \sqrt{3}) = \pi/12$ or $x = \tan^{-1}(2 + \sqrt{3}) = 5\pi/12$.

答案 06/08

Correct answer and working. Since $\sin(2x)$ has period π , $\operatorname{cosec}(2x)$ also has period π . It is undefined when $\sin 2x = 0$, so $x = k\pi/2$ for integer k .

答案 07/08

Correct answer and working. $R = \sqrt{8^2 + 15^2} = 17$, so the range is $[-17, 17]$.

答案 08/08

Correct answer and working. The left side has maximum $\sqrt{3+4}=5$, so it cannot equal 6.

6. 自学检查清单 | Self-study checklist (共 10 项)

- ☐ 能准确说明 六个三角函数与定义域
- ☐ 能准确说明 恒等式的选择来自目标形式
- ☐ 能准确说明 和差角与二倍角
- ☐ 能准确说明 $R\sin(x+\alpha)$ 或 $R\cos(x-\alpha)$
- ☐ 能准确说明 三角方程必须列全区间解
- ☐ 能准确说明 用辅助角求范围与方程根
- ☐ 能准确说明 解方程时避免除掉可能为零的因子
- ☐ 能在没有答案提示时完成本单元全部练习并保留关键步骤
- ☐ 能检查定义域、区间、符号、精度或几何意义，而不是只抄计算器数值
- ☐ 能用英文说明所选公式或方法为何适用于当前条件

3.4 | 微分 | Differentiation

1. 单元导读 | Unit introduction

本单元严格对应官方 3.4。学习时先建立概念与条件，再推导公式，随后通过完整 worked examples 把方法转化为可复现步骤。P3 的得分关键是：选择正确结构、展示 decisive working、保持 exact values，并检查定义域、区间、参数或几何意义。

与国内课程衔接 | Connection. 国内高中教材已提供相近的代数与微积分基础；Cambridge 更强调多步骤结构题、英文命令词、前后小问的 hence 连接，以及对限制条件的显式说明。

2. 知识点精讲 | Detailed teaching (共 7 个知识点)

知识点 01/07 | 新函数导数与链式法则 | Derivatives and the chain rule

$d(e)/dx = e \, du/dx$; $d(\ln u)/dx = (1/u)du/dx$; $d(\tan u)/dx = \sec u \, du/dx$; $d(\tan u)/dx = (1/(1+u))du/dx$ 。链式法则的内层导数不能遗漏。

English explanation and method. Example. $y = e^{(3x)}$ gives $dy/dx = 6x \, e^{(3x)}$. For $y = \ln(2x-1)$, $dy/dx = 2/(2x-1)$, with original domain $x > 1/2$.

知识点 02/07 | 乘积、商与结构识别 | Product and quotient rules

乘积: $(uv)' = u'v + uv'$; 商: $(u/v)' = (vu' - uv')/v$ 。先识别整体结构，再在草稿中分别写 u 、 v 、 u' 、 v' 。

English explanation and method. Worked example. $y = x \ln x$ gives $y' = 2x \ln x + x$. For $y = e/(1+x)$, $y' = [e(1+x) - e]/(1+x) = xe/(1+x)$. Simplification after correct differentiation often reveals stationary points.



知识点 03/07 | 隐函数微分 | Implicit differentiation

当 x 、 y 混在同一关系中，把 y 看成 $y(x)$ 。对 y 微分得 $2y \, dy/dx$ ；对 xy 用乘积法则得 $y + x \, dy/dx$ 。最后集中 dy/dx 。

English explanation and method. Example. $x+y=xy+7$ gives $2x+2y \, y' = y+x \, y'$. Hence $y' (2y-x) = y-2x$ and $y' = (y-2x)/(2y-x)$. At a point, substitute coordinates only after obtaining the derivative.

知识点 04/07 | 参数方程微分 | Parametric differentiation

若 $x=f(t)$, $y=g(t)$ ，则 $dy/dx = (dy/dt)/(dx/dt)$ ，前提 $dx/dt \neq 0$ 。切线斜率来自 dy/dx ；法线斜率是 $-1/(dy/dx)$ ，但水平/竖直情形须单独处理。

English explanation and method. Example. $x=t+1, y=t-t$. Then $dy/dx=(3t-1)/(2t)$. At $t=1$, the point is $(2,0)$, tangent gradient is 1, so tangent is $y=x-2$.

知识点 05/07 | 切线、法线与定义域检查 | Tangents and normals

得到梯度 m 后使用 $y-y=m(x-x)$ 。法线梯度 $-1/m$ 只适用于 m 非零有限；若 tangent horizontal, normal vertical。

English explanation and method. In a full solution, show the point, the derivative, the parameter or coordinate substituted, and the line equation. A bare line equation hides the method and risks losing method marks.

知识点 06/07 | 驻点与曲线性质的证据链 | Stationary-point classification

先解 $dy/dx=0$ 得候选点，再用 dy/dx 的符号或一阶导数变号分类。 $dy/dx>0$ 为局部极小， <0 为局部极大；等于 0 时结论不确定，必须另查符号或更高阶行为。

English explanation and method. Example. For $f(x)=x-3x$, $f' = 3x-3$ gives $x=\pm 1$. Since $f'' = 6x$, $x=-1$ gives $f'' < 0$ and a local maximum $f(-1)=2$; $x=1$ gives $f'' > 0$ and a local minimum $f(1)=-2$.

知识点 07/07 | 相关变化率先画变量关系 | Related rates

多个量随时间变化时，先写它们之间的几何或物理关系，再对 t 微分。半径 r 、体积 V 都是 t 的函数，因此 $dV/dt=(dV/dr)(dr/dt)$ 。

English explanation and method. Example. A sphere has $V=4\pi r^3/3$. If $dr/dt=0.2$ at $r=5$, then $dV/dt=4\pi r^2 dr/dt=4\pi \cdot 25 \cdot 0.2=20\pi$. Units are cubic units per unit time, not length per time.

3. 核心词汇表 | Core vocabulary (共 10 词)

English term	中文准确口径	Exam-ready meaning
chain rule	链式法则	differentiation of a composite function.
product rule	乘积法则	differentiation rule for a product.
quotient rule	商法则	differentiation rule for a quotient.
implicit differentiation	隐函数微分	differentiation without first isolating y .
parametric equation	参数方程	coordinates expressed using a third variable.
tangent	切线	a line with the curve gradient at a point.
normal	法线	a line perpendicular to the tangent.
stationary point	驻点	a point where the first derivative is zero.
composite function	复合函数	a function applied inside another function.
rate of change	变化率	a derivative describing change of one variable with another.

4. 单元练习 | Unit practice (共 8 题)

练习 01/08

Question. Differentiate $y=e^{(3x)}\ln x$. [4]

练习 02/08

Question. Differentiate $y=\tan^{-1}(2x-1)$. [3]

练习 03/08

Question. Given $x+y=xy+7$, find dy/dx . [5]

练习 04/08

Question. $x=t+1, y=t-t$. Find the tangent at $t=1$. [5]

练习 05/08

Question. Find the normal to $y=x\ln x$ at $x=1$. [5]

练习 06/08

Question. Explain why $dy/dx=(dy/dt)/(dx/dt)$ requires $dx/dt \neq 0$. [2]

练习 07/08

Question. Classify the stationary points of $f(x)=x-3x$. [6]

练习 08/08

Question. A sphere radius increases at 0.2 cm/s . Find dV/dt at $r=5 \text{ cm}$. [4]

5. 练习答案与解析 | Answers and explanations (共 8 题)

答案 01/08

Correct answer and working. Product and chain rules: $y' = 6xe^{(3x)}\ln x + e^{(3x)}/x = e^{(3x)}(6x\ln x + 1/x)$.

答案 02/08

Correct answer and working. Let $u=2x-1$. Then $y' = u'/(1+u) = 2/[1+(2x-1)]$.

答案 03/08

Correct answer and working. $2x+2yy' = y+xy'$. Hence $y'(2y-x)=y-2x$ and $dy/dx=(y-2x)/(2y-x)$.

答案 04/08

Correct answer and working. $dy/dx=(3t-1)/(2t)$. At $t=1$, gradient=1 and point=(2,0), so tangent $y=x-2$.

答案 05/08

Correct answer and working. $y' = 2x\ln x + x$, so at $x=1$, $y=0$ and tangent gradient=1. Normal gradient=-1; equation $y=-x+1$.

答案 06/08

Correct answer and working. The formula divides by dx/dt . If $dx/dt=0$ it is undefined and the curve may have a vertical tangent; that case must be examined separately.

答案 07/08

Correct answer and working. $f' = 3x-3=0$ gives $x=\pm 1$. $f'' = 6x$: at -1 it is negative, so $(-1,2)$ is a local maximum; at 1 positive, so $(1,-2)$ is a local minimum.

答案 08/08

Correct answer and working. $V=4\pi r^3/3$, so $dV/dt=4\pi r dr/dt=4\pi \cdot 25 \cdot 0.2=20\pi \text{ cm}^3/\text{s}$.

6. 自学检查清单 | Self-study checklist (共 10 项)

☐ 能准确说明 新函数导数与链式法则

- ☐ 能准确说明 乘积、商与结构识别
- ☐ 能准确说明 隐函数微分
- ☐ 能准确说明 参数方程微分
- ☐ 能准确说明 切线、法线与定义域检查
- ☐ 能准确说明 驻点与曲线性质的证据链
- ☐ 能准确说明 相关变化率先画变量关系
- ☐ 能在没有答案提示时完成本单元全部练习并保留关键步骤
- ☐ 能检查定义域、区间、符号、精度或几何意义，而不是只抄计算器数值
- ☐ 能用英文说明所选公式或方法为何适用于当前条件

3.5 | 积分 | Integration

1. 单元导读 | Unit introduction

本单元严格对应官方 3.5。学习时先建立概念与条件，再推导公式，随后通过完整 worked examples 把方法转化为可复现步骤。P3 的得分关键是：选择正确结构、展示 decisive working、保持 exact values，并检查定义域、区间、参数或几何意义。

与国内课程衔接 | Connection. 国内高中教材已提供相近的代数与微积分基础；Cambridge 更强调多步骤结构题、英文命令词、前后小问的 hence 连接，以及对限制条件的显式说明。

2. 知识点精讲 | Detailed teaching (共 8 个知识点)

知识点 01/08 | 基本积分是反向链式法则 | Standard integrals

$\int e^{ax+b} dx = e^{ax+b}/a + C$; $\int 1/(ax+b) dx = (1/a) \ln|ax+b| + C$; $\int \sin(ax+b) dx = -\cos(ax+b)/a + C$; $\int \sec(ax+b) dx = \tan(ax+b)/a + C$ 。

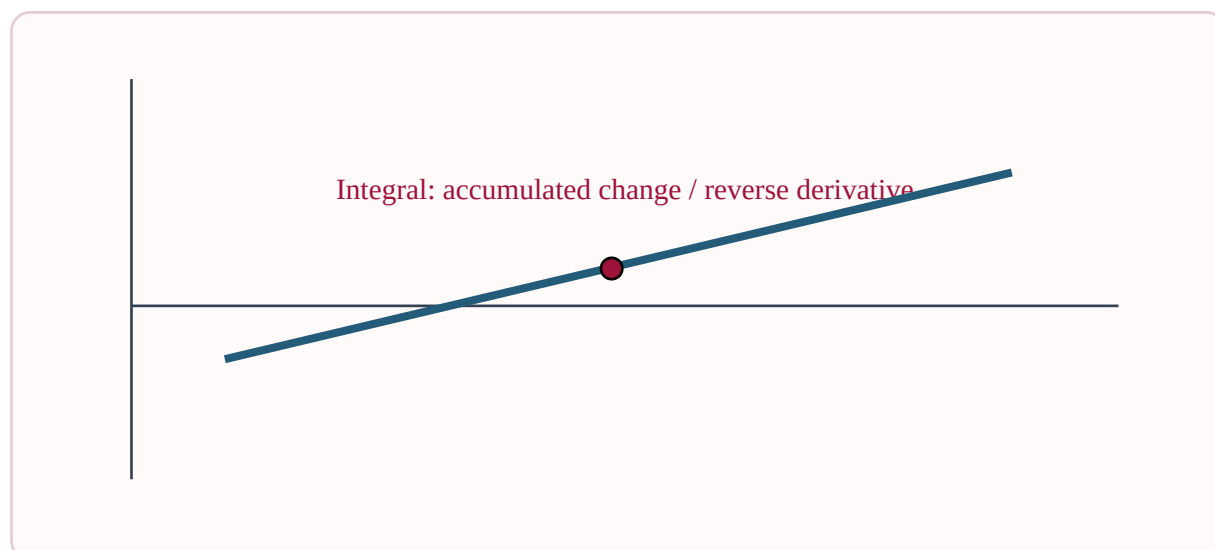
English explanation and method. Differentiate the answer mentally: the inner coefficient a must cancel. The absolute value in $\ln|ax+b|$ preserves validity on intervals where $ax+b$ is negative; omit C only for a definite integral.

知识点 02/08 | 看到 $f'(x)/f(x)$ 就想到 $\ln$ | Recognising f' -prime over f

若 integrand = $k f'(x)/f(x)$ ，积分为 $k \ln|f(x)| + C$ 。

识别的关键是分子与分母导数成常数倍，而不是外观是否“像分式”。

English explanation and method. Example. $\int (6x)/(3x+5) dx = \ln(3x+5) + C$ because the denominator derivative is $6x$. But $\int x/(3x+5) dx = (1/6) \ln(3x+5) + C$.



知识点 03/08 | 三角恒等式先把被积式变成熟悉形式 | Trig integration

$\sin x = (1 - \cos 2x)/2$, $\cos x = (1 + \cos 2x)/2$. $\tan x = \sin x / \cos x$ 可视为 $-f'/f$ ，因此 $\int \tan x dx = -\ln|\cos x| + C$ 。

English explanation and method. Example. $\int \cos(2x) dx = \int (1 + \cos 4x)/2 dx = x/2 + \sin 4x/8 + C$. The factor 4 in the inner angle creates the denominator 8.

知识点 04/08 | 分部积分：选择谁变简单 | Integration by parts

公式 $\int u dv = uv - \int v du$ 来自乘积法则。通常令 algebraic/log/inverse-trig 为 u ，容易积分的部分为 dv 。

English explanation and method. Worked example. $\int x \, \text{e}^x \, \text{d}x$: choose $u=x, \text{d}v=\text{e}^x \, \text{d}x$, so $\text{d}u=\text{d}x, v=\text{e}^x$. Hence $\int x \, \text{e}^x \, \text{d}x = x\text{e}^x - \int \text{e}^x \, \text{d}x = \text{e}^x(x-1) + C$. For $\int \ln x \, \text{d}x$, write $\ln x = 1 \cdot \ln x$ and choose $u=\ln x$.

知识点 05/08 | 给定代换要同时换 $\text{d}x$ 与界 | Substitution

设 $u=g(x)$, 则 $\text{d}u=g'(x)\text{d}x$. 定积分可把 x 界转换为 u 界, 并全程使用 u ; 或求出 x 的原函数再用原界, 不能混用。
English explanation and method. Example. For $\int 2x(x+1)\text{d}x$, let $u=x+1, \text{d}u=2x \, \text{d}x$. Limits: $x=0 \rightarrow u=1, x=1 \rightarrow u=2$. Integral $= \int u \, \text{d}u = [u^2/2] = 15/4$.

知识点 06/08 | 部分分式积分把代数连接到积分 | Partial fractions in integration

先按 3.1 分解, 再逐项积分。线性分母产生 logarithm; 重因式的 $1/(ax+b)$ 产生负一次幂项。
English explanation and method. Always verify the decomposition before integrating. In definite integrals, check denominator zeros in the interval: crossing a singularity means the ordinary definite integral is not valid.

知识点 07/08 | 定积分的换元与方向 | Definite substitution

若 u 随 x 递减, 新上下限的顺序会自然反转; 不要擅自交换界而忘记负号。完成换元后 integrand、differential 和 limits 必须全部属于同一变量。
English explanation and method. Example. With $u=1-x$ in $\int f(1-x)\text{d}x$, the limits become $u=1$ to $u=0$ and $\text{d}x=-\text{d}u$. Thus the two reversals cancel, giving $\int f(u)\text{d}u$.

知识点 08/08 | 积分结果的微分核验 | Checking by differentiation

不定积分最可靠的检查是把答案微分回 integrand。分部积分常错在负号, 三角积分常错在内层系数, 部分分式常错在系数; 微分检查能分别暴露这些错误。
English explanation and method. For a definite integral, also estimate sign and size from the graph. If a positive integrand on $[0,1]$ produces a negative answer, the algebra or limits must be wrong even if the calculator accepts the expression.

3. 核心词汇表 | Core vocabulary (共 10 词)

English term	中文准确口径	Exam-ready meaning
antiderivative	原函数	a function whose derivative is the integrand.
constant of integration	积分常数	the arbitrary constant in an indefinite integral.
definite integral	定积分	an accumulated signed value between limits.
integration by parts	分部积分	integration derived from the product rule.
substitution	换元	changing variable to simplify an integral.
integrand	被积函数	the function being integrated.
limit	积分限	an endpoint of a definite integral.
reverse chain rule	反向链式法则	recognising a derivative inside a composite.
singularity	奇点	a point where the integrand is undefined.
exact value	精确值	a value retained using logs, pi or radicals.

4. 单元练习 | Unit practice (共 8 题)

练习 01/08

Question. Evaluate $\int (6x)/(3x+5) \, \text{d}x$. [3]

练习 02/08

Question. Evaluate $\int \cos(2x) \, dx$. [4]

练习 03/08

Question. Evaluate $\int x e^x \, dx$. [4]

练习 04/08

Question. Evaluate $\int 2x(x+1) \, dx$. [5]

练习 05/08

Question. Evaluate $\int \ln x \, dx$. [4]

练习 06/08

Question. Integrate $(7x+5)/[(x+1)(x+2)]$. [5]

练习 07/08

Question. Evaluate $\int (1-x) \, dx$ using $u=1-x$. [4]

练习 08/08

Question. Verify by differentiation that $e^x(x-1)$ is an antiderivative of xe^x . [2]

5. 练习答案与解析 | Answers and explanations (共 8 题)

答案 01/08

Correct answer and working. The numerator is the denominator derivative, so the integral is $\ln(3x+5)+C$.

答案 02/08

Correct answer and working. $\cos(2x) = (1+\cos 4x)/2$. Hence $\text{integral} = x/2 + \sin 4x/8 + C$.

答案 03/08

Correct answer and working. By parts with $u=x, dv=e^x \, dx$: $xe^x - \int e^x \, dx = e^x(x-1) + C$.

答案 04/08

Correct answer and working. Let $u=x+1, du=2x \, dx$; limits 1 to 2. $\text{Integral} = [u/4] = (16-1)/4 = 15/4$.

答案 05/08

Correct answer and working. Write $\ln x \cdot 1$. By parts $u=\ln x, dv=dx$: $x \ln x - \int 1 \, dx = x \ln x - x + C$.

答案 06/08

Correct answer and working. Decompose as $-2/(x+1) + 9/(x+2)$. $\text{Integral} = -2 \ln|x+1| + 9 \ln|x+2| + C$.

答案 07/08

Correct answer and working. $u=1-x, dx=-du$; limits 1 to 0. $\text{Integral} = - \int u \, du = \int u \, du = 1/3$.

答案 08/08

Correct answer and working. Derivative $= e^x(x-1) + e^x = xe^x$, so the antiderivative is verified.

6. 自学检查清单 | Self-study checklist (共 11 项)

- ☐ 能准确说明 基本积分是反向链式法则
- ☐ 能准确说明 看到 $f'(x)/f(x)$ 就想到 $\ln$
- ☐ 能准确说明 三角恒等式先把被积式变成熟悉形式
- ☐ 能准确说明 分部积分: 选择谁变简单
- ☐ 能准确说明 给定代换要同时换 dx 与界
- ☐ 能准确说明 部分分式积分把代数连接到积分
- ☐ 能准确说明 定积分的换元与方向
- ☐ 能准确说明 积分结果的微分核验
- ☐ 能在没有答案提示时完成本单元全部练习并保留关键步骤
- ☐ 能检查定义域、区间、符号、精度或几何意义, 而不是只抄计算器数值
- ☐ 能用英文说明所选公式或方法为何适用于当前条件

3.6 | 方程的数值解 | Numerical solution of equations

1. 单元导读 | Unit introduction

本单元严格对应官方 3.6。学习时先建立概念与条件，再推导公式，随后通过完整 worked examples 把方法转化为可复现步骤。P3 的得分关键是：选择正确结构、展示 decisive working、保持 exact values，并检查定义域、区间、参数或几何意义。

与国内课程衔接 | Connection. 国内高中教材已提供相近的代数与微积分基础；Cambridge 更强调多步骤结构题、英文命令词、前后小问的 hence 连接，以及对限制条件的显式说明。

2. 知识点精讲 | Detailed teaching (共 6 个知识点)

知识点 01/06 | 符号变化定位根 | Locating a root

若连续函数 f 在 $[a,b]$ 上满足

$f(a)f(b)<0$ ，则区间内至少有一根。必须写出两端函数值及异号结论；只有“图上看见”通常不够。

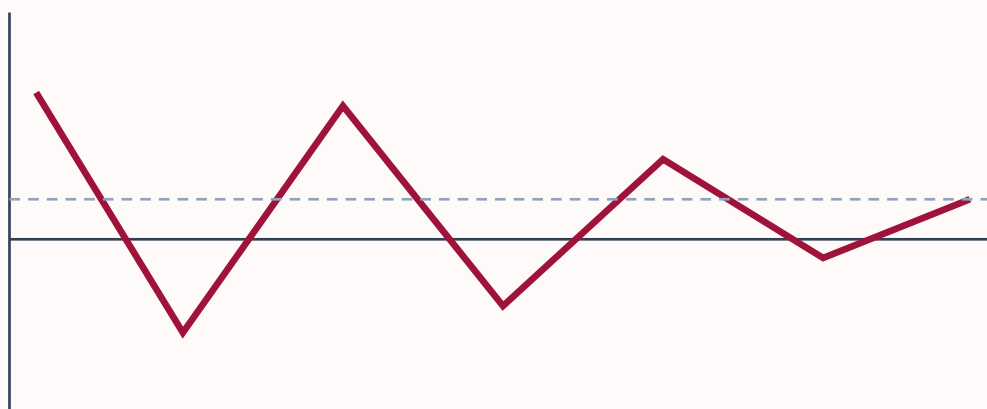
English explanation and method. A sign change proves at least one crossing for a continuous function, not exactly one. Uniqueness needs extra information such as monotonicity or a graph.

知识点 02/06 | 迭代式来自原方程重排 | Fixed-point iteration

把 $f(x)=0$ 重排为 $x=F(x)$ ，再定义

$x=F(x)$ 。不同重排可有不同收敛表现；题目不要求证明收敛条件，但要能识别数列可能发散或振荡。

English explanation and method. Worked example. For $x+x-3=0$, use $x=(3-x)^{1/3}$. Starting $x=1$ gives $x \approx 1.2599$, $x \approx 1.2038$, $x \approx 1.2170$; values settle near the root.



successive approximations should settle towards a fixed point

知识点 03/06 | 按指定精度停止 | Accuracy and stopping

若要求 3 decimal places，需确认根在相同三位小数的舍入区间内。连续两个迭代值相同到 3 d.p. 是常见证据，但更严谨可检查 f 在相邻舍入边界的符号。

English explanation and method. Do not round every intermediate value. Store full calculator precision and report only at the end; early rounding can make a sequence appear falsely stable.

知识点 04/06 | 图像解释 $x=F(x)$ 的迭代 | Cobweb interpretation

迭代在 $y=F(x)$ 与 $y=x$ 之间来回投影。靠近交点时若映射把误差缩小，数列收敛；若误差放大则发散。考纲不要求形式化收敛判据，但理解图形有助于选择初值。

English explanation and method. A numerical answer is not an exact root. State it with the requested accuracy and, when asked, show enough iterations or a sign-change interval to justify that accuracy.

知识点 05/06 | 唯一性与单调性 | Uniqueness of a located root

异号只保证至少一根。若还可证明 $f'(x) > 0$ 在区间恒成立，则 f 严格递增，因此至多一根；结合异号便得到恰有一根。

English explanation and method. Example. $f(x)=x+x-3$ has $f'(x)=3x+1 > 0$ for all x . Since $f(1) < 0 < f(2)$, it has exactly one root in $(1,2)$.

知识点 06/06 | 迭代表格的规范呈现 | Presenting iteration

表格至少列 n 与 x ，并保留足够小数。若题目给起点和迭代式，必须按索引正确计算： $x=F(x)$ ，不是把 x 误标为 x_0 。

English explanation and method. When the requested answer is r to 3 d.p., report r only after the displayed iterates have stabilised sufficiently. If the sequence fails to settle, state that this rearrangement is unsuitable rather than inventing a root.

3. 核心词汇表 | Core vocabulary (共 10 词)

English term	中文准确口径	Exam-ready meaning
root	根	a value satisfying an equation.
sign change	符号变化	opposite signs at two endpoints.
continuous function	连续函数	a function without a break on an interval.
iteration	迭代	repeated use of a formula to form approximations.
initial value	初值	the first approximation x_0 .
convergence	收敛	approximations approaching a limiting value.
divergence	发散	approximations failing to settle to a root.
fixed point	不动点	a value satisfying $x=F(x)$.
prescribed accuracy	指定精度	the rounding or error requirement stated.
approximation	近似值	a numerical value close to the exact result.

4. 单元练习 | Unit practice (共 8 题)

练习 01/08

Question. Show that $x+x-3=0$ has a root between 1 and 2. [3]

练习 02/08

Question. Use $x_{n+1}=(3-x_n)^{1/3}$, $x_0=1$, to find x_1 and x_2 to 4 d.p. [3]

练习 03/08

Question. Explain how $x_{n+1}=(3-x_n)^{1/3}$ relates to $x+x-3=0$. [2]

练习 04/08

Question. A sequence alternates 1.1, 2.8, 1.0, 3.0,... . What does this indicate? [2]

练习 05/08

Question. State a reliable way to justify a root correct to 2 d.p. [3]

练习 06/08

Question. Why should full calculator precision be retained during iteration? [2]

练习 07/08

Question. Show $x+x-3=0$ has exactly one real root. [4]

练习 08/08

Question. What is x_1 if $x_0=1.4$ and $x_{n+1}=\cos x_n$? [2]

5. 练习答案与解析 | Answers and explanations (共 8 题)

答案 01/08

Correct answer and working. $f(1)=-1$ and $f(2)=7$. Since f is continuous and the signs differ, at least one root lies in $(1,2)$.

答案 02/08

Correct answer and working. $x_1=2^{1/3}=1.2599$. $x_2=(3-1.259921\dots)^{1/3}=1.2038$ to 4 d.p.

答案 03/08

Correct answer and working. At a fixed point $x=(3-x)^{1/3}$. Cubing gives $x=3-x$, which rearranges to $x+x-3=0$.

答案 04/08

Correct answer and working. The approximations do not settle towards one value; the chosen iteration appears to diverge or oscillate, so it should not be used to claim a root.

答案 05/08

Correct answer and working. Check f at the two rounding boundaries around the stated value, e.g. $r-0.005$ and $r+0.005$. If f is continuous and changes sign, the root lies inside the interval that rounds to r .

答案 06/08

Correct answer and working. Rounding each iterate changes the input to the next step and can accumulate error or create false stability. Round only the reported answer.

答案 07/08

Correct answer and working. $f(1)<0<f(2)$ gives at least one root. $f'=3x+1>0$ for every real x , so f is strictly increasing and has at most one root; hence exactly one.

答案 08/08

Correct answer and working. $x_1=\cos(1.4) \approx 0.16997$, using radians.

6. 自学检查清单 | Self-study checklist (共 9 项)

- ☐ 能准确说明 符号变化定位根
- ☐ 能准确说明 迭代式来自原方程重排

- ☐ 能准确说明 按指定精度停止
- ☐ 能准确说明 图像解释 $x=F(x)$ 的迭代
- ☐ 能准确说明 唯一性与单调性
- ☐ 能准确说明 迭代表格的规范呈现
- ☐ 能在没有答案提示时完成本单元全部练习并保留关键步骤
- ☐ 能检查定义域、区间、符号、精度或几何意义，而不是只抄计算器数值
- ☐ 能用英文说明所选公式或方法为何适用于当前条件

3.7 | 向量 | Vectors

1. 单元导读 | Unit introduction

本单元严格对应官方 3.7。学习时先建立概念与条件，再推导公式，随后通过完整 worked examples 把方法转化为可复现步骤。P3 的得分关键是：选择正确结构、展示 decisive working、保持 exact values，并检查定义域、区间、参数或几何意义。

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2. 知识点精讲 | Detailed teaching (共 7 个知识点)

知识点 01/07 | 位置、位移、模与单位向量 | Vector foundations

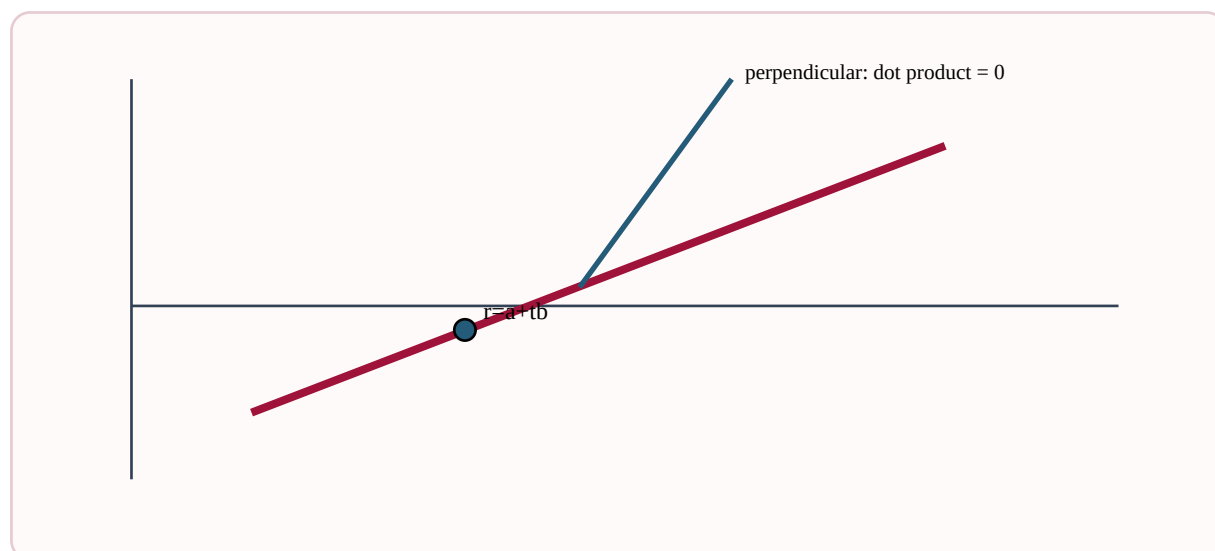
点 A 的 position vector 是 $\vec{OA}$ ； $\vec{AB} = \vec{OB} - \vec{OA}$ 。若 $\mathbf{a} = (x, y, z)$ ， $|\mathbf{a}| = \sqrt{x^2 + y^2 + z^2}$ ，同方向单位向量为 $\mathbf{a}/|\mathbf{a}|$ 。中点位置向量为 $(\mathbf{a} + \mathbf{b})/2$ 。

English explanation and method. Example. $\mathbf{A} = (1, -2, 3)$, $\mathbf{B} = (5, 1, -1)$. Then $\vec{AB} = (4, 3, -4)$, $|\vec{AB}| = \sqrt{41}$, and the unit vector from A to B is $(4, 3, -4)/\sqrt{41}$.

知识点 02/07 | 直线 $\mathbf{r} = \mathbf{a} + t\mathbf{b}$ 的每个符号 | Vector equation of a line

$\mathbf{a}$ 是线上一个固定点的位置向量； $\mathbf{b}$ 是非零方向向量； t 是实参数。由两点 A、B 可取 $\mathbf{r} = \mathbf{a} + t(\mathbf{b} - \mathbf{a})$ 。方向向量可乘任意非零常数而不改变直线。

English explanation and method. To test whether a point lies on the line, equate all three components and require one common parameter value. Matching only two coordinates is insufficient in 3D.



知识点 03/07 | 平行、相交与异面 | Parallel, intersecting and skew lines

方向向量成比例则平行。否则联立 $\mathbf{a} + t\mathbf{b} = \mathbf{c} + s\mathbf{u}$ ；若三分量存在同一组 t, s ，则相交；若方程不一致，则两线异面。

English explanation and method. Worked example. Solve two component equations first, then check the third. The third-coordinate check is decisive: in a plane non-parallel lines meet, but in 3D they may be skew.

知识点 04/07 | 点积与夹角 | Scalar product

$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3 = |\mathbf{a}||\mathbf{b}|\cos\theta$ 。点积为 0 表示非零向量垂直。直线夹角通常取 acute angle，因此必要时使用 $|\mathbf{a} \cdot \mathbf{b}|$ 。

English explanation and method. Example. $a=(1,2,-1)$, $b=(2,-1,0)$: $a \cdot b=2-2+0=0$, so the directions are perpendicular. For an angle, $\cos \theta=(a \cdot b)/(|a||b|)$, then check whether the question wants the acute angle.

知识点 05/07 | 从点到直线的垂足 | Foot of a perpendicular

线上点 $H=a+tb$ 。垂足条件 PH 与方向 b 垂直，所以 $(H-P) \cdot b=0$ ；解 t 后求 H 。

English explanation and method. This single scalar equation is safer than inventing a perpendicular line. Shortest distance is then $|PH|$. The syllabus does not require shortest distance between two skew lines or a vector product.

知识点 06/07 | 两直线相交的联立顺序 | Solving line intersections

从两个分量解 t, s ，再把所得参数代入第三分量检查。若第三式不成立，不能说“计算错了”后强行求交点；这正是 skew 的判据。

English explanation and method. Example. If x - and y -components give $t=2, s=-1$ but z -components then give $5=7$, the lines are non-parallel yet have no common point, so they are skew.

知识点 07/07 | 向量几何中的比例与内分点 | Points dividing a segment

若 P 在 AB 上且 $AP:PB=m:n$ ，则 $OP=(n \cdot OA+m \cdot OB)/(m+n)$ 。更基础地，也可写 $OP=OA+\lambda AB$ ，并由比例求 $\lambda=m/(m+n)$ ，避免死记公式。

English explanation and method. For a midpoint $\lambda=1/2$, so $OP=(OA+OB)/2$. The parameter form shows why the point lies on the segment and generalises naturally to other ratios.

3. 核心词汇表 | Core vocabulary (共 10 词)

English term	中文准确口径	Exam-ready meaning
position vector	位置向量	a vector from the origin to a point.
displacement vector	位移向量	the change from one position to another.
magnitude	模长	the length of a vector.
unit vector	单位向量	a vector of length one.
direction vector	方向向量	a non-zero vector parallel to a line.
parameter	参数	a scalar locating points on a line.
parallel lines	平行线	lines with proportional direction vectors.
skew lines	异面直线	non-parallel 3D lines that do not intersect.
scalar product	点积	a dot product equal to $ a b \cos \theta$.
foot of perpendicular	垂足	the point where a perpendicular meets a line.

4. 单元练习 | Unit practice (共 8 题)

练习 01/08

Question. $A=(1,-2,3)$, $B=(5,1,-1)$. Find AB and its magnitude. [3]

练习 02/08

Question. Find a vector equation of the line through $A=(1,2,0)$ and $B=(3,-1,4)$. [3]

练习 03/08

Question. Show that $(1,2,-1)$ and $(2,-1,0)$ are perpendicular. [2]

练习 04/08

Question. Find the acute angle between $a=(1,1,0)$ and $b=(1,0,1)$. [4]

练习 05/08

Question. Explain the difference between intersecting and skew lines. [3]

练习 06/08

Question. $P=(1,0,0)$, line $r=(0,1,0)+t(1,1,0)$. Find the perpendicular foot H. [5]

练习 07/08

Question. Find the point dividing $A=(1,2,3)$, $B=(6,-3,8)$ in ratio $AP:PB=2:3$. [4]

练习 08/08

Question. Two non-parallel 3D lines give inconsistent component equations. Classify them. [2]

5. 练习答案与解析 | Answers and explanations (共 8 题)

答案 01/08

Correct answer and working. $AB=B-A=(4,3,-4)$, and $|AB|=\sqrt{(16+9+16)}=\sqrt{41}$.

答案 02/08

Correct answer and working. Direction $AB=(2,-3,4)$. Thus $r=(1,2,0)+t(2,-3,4)$, t real.

答案 03/08

Correct answer and working. Dot product $=1 \cdot 2 + 2 \cdot (-1) + (-1) \cdot 0 = 0$. Since both are non-zero, they are perpendicular.

答案 04/08

Correct answer and working. $a \cdot b = 1$, $|a|=|b|=\sqrt{2}$, so $\cos \theta = 1/2$ and $\theta = 60^\circ$.

答案 05/08

Correct answer and working. Intersecting lines have a common point, so their component equations share one pair of parameter values. Skew lines are non-parallel 3D lines with no common point.

答案 06/08

Correct answer and working. $H=(t,1+t,0)$. $PH=(t-1,1+t,0)$. Set $PH \cdot (1,1,0)=0$: $t-1+1+t=2t=0$, so $t=0$ and $H=(0,1,0)$.

答案 07/08

Correct answer and working. $OP=OA+(2/5)AB=(1,2,3)+(2/5)(5,-5,5)=(3,0,5)$.

答案 08/08

Correct answer and working. They do not intersect and are not parallel, so they are skew.

6. 自学检查清单 | Self-study checklist (共 10 项)

- ☐ 能准确说明 位置、位移、模与单位向量
- ☐ 能准确说明 直线 $r=a+tb$ 的每个符号

- ☐ 能准确说明 平行、相交与异面
- ☐ 能准确说明 点积与夹角
- ☐ 能准确说明 从点到直线的垂足
- ☐ 能准确说明 两直线相交的联立顺序
- ☐ 能准确说明 向量几何中的比例与内分点
- ☐ 能在没有答案提示时完成本单元全部练习并保留关键步骤
- ☐ 能检查定义域、区间、符号、精度或几何意义，而不是只抄计算器数值
- ☐ 能用英文说明所选公式或方法为何适用于当前条件

3.8 | 微分方程 | Differential equations

1. 单元导读 | Unit introduction

本单元严格对应官方 3.8。学习时先建立概念与条件，再推导公式，随后通过完整 worked examples 把方法转化为可复现步骤。P3 的得分关键是：选择正确结构、展示 decisive working、保持 exact values，并检查定义域、区间、参数或几何意义。

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2. 知识点精讲 | Detailed teaching (共 7 个知识点)

知识点 01/07 | 把文字变化率写成方程 | Forming a differential equation

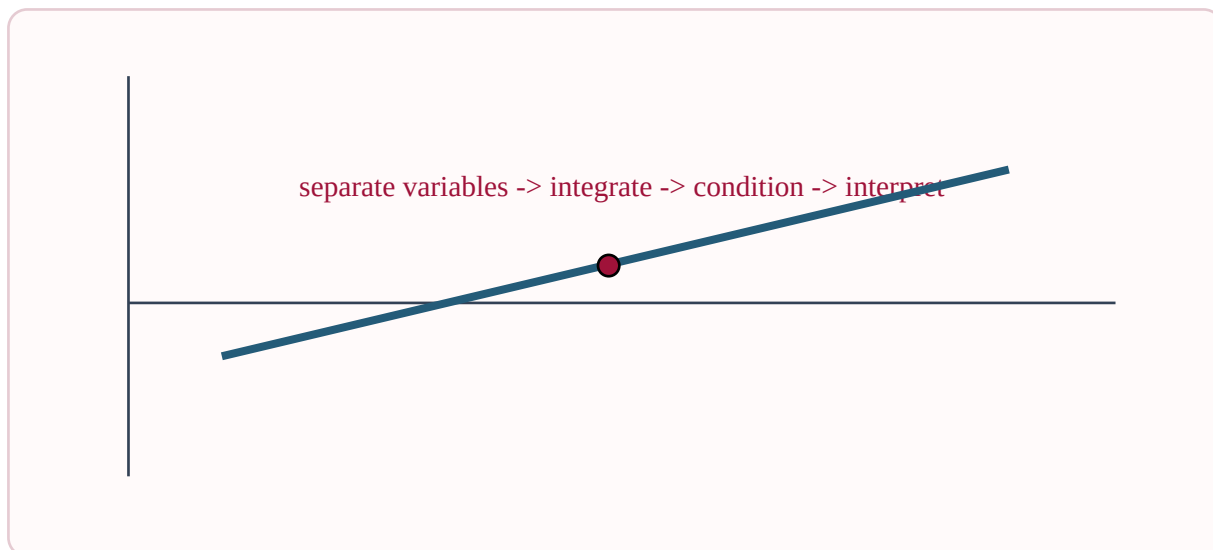
“rate of change of y is proportional to ...” 写为 $dy/dx=k(\dots)$ ；increases 用 $k>0$ ，decreases 可写负号或让 $k<0$ 。变量、单位和初始时刻必须明确。

English explanation and method. Example: “The rate of decrease of mass M is proportional to M ” becomes $dM/dt=-kM$ with $k>0$. The minus sign records decrease; it is not added after solving.

知识点 02/07 | 分离变量的逻辑 | Separating variables

目标是把所有 y 因子与 dy 放一边，把 x 因子与 dx 放另一边，再两边积分。除以可能为零的表达式时，要检查是否丢失 equilibrium solution。

English explanation and method. Example. $dy/dx=xy$ gives $(1/y)dy=x dx$ for $y \neq 0$. Hence $\ln|y|=x/2+C$, so $y=Ae^{(x/2)}$. Also $y=0$ is a solution and should be considered if the initial condition permits it.



知识点 03/07 | 积分常数与初始条件 | General and particular solutions

两边积分只需要一个独立常数 C 。初始条件如 $y=3$ when $x=0$ 用来求 C 或 A ，从 general solution 变成 particular solution。

English explanation and method. For $dy/dx=2x(y+1)$, integration gives $\ln|y+1|=x+C$, hence $y+1=Ae^x(x)$. If $y(0)=2$, then $A=3$ and $y=3e^x-1$.

知识点 04/07 | 解释模型而不是只给函数 | Interpreting a model

解得函数后要回答题目中的时间、数量或极限。检查初值、符号、单位以及长期行为是否符合情境。

English explanation and method. If a population model gives $P=Ce^{kt}$, $k>0$ implies growth and $k<0$ decay. A mathematically valid negative population may be outside the model's meaningful domain; state such restrictions.

知识点 05/07 | 含 f'/f 或部分分式的分离积分 | Using integration techniques

P3 微分方程可能需要 3.5 的部分分式、对数或代换。先正确分离，再选择积分技术；不要在 y 仍与 x 混合时直接积分。

English explanation and method. Example. $dy/dx=y(1-y)$ gives $dy/[y(1-y)]=dx$. Since $1/[y(1-y)]=1/y+1/(1-y)$, integration gives $\ln|y|-\ln|1-y|=x+C$. Then solve for y only if the question requires an explicit form.

知识点 06/07 | 平衡解与长期行为 | Equilibria and long-run behaviour

令 $dy/dx=0$ 可找 constant equilibrium solutions。对 logistic 型 $dy/dt=ky(L-y)$ ，平衡为 $y=0$ 与 $y=L$ ；在 $0<y<L$ 时增长，在 $y>L$ 时下降，因此 L 是稳定水平。

English explanation and method. This interpretation comes from the sign of the derivative, not from solving first. A phase-line sign analysis can predict whether a model approaches or moves away from an equilibrium.

知识点 07/07 | 从隐式解到显式解 | Rearranging a differential-equation solution

部分分式积分后常得到 $\ln|y|-\ln|L-y|=kt+C$ ，即 $\ln|y/(L-y)|=kt+C$ 。指数化后 $y/(L-y)=Ae^{kt}$ ，再解 $y=LAe^{kt}/[1+Ae^{kt}]$ 。

English explanation and method. Keep the integration constant flexible: after exponentiation, e^C becomes a non-zero multiplicative constant A . An initial condition determines its value and sign.

3. 核心词汇表 | Core vocabulary (共 10 词)

English term	中文准确口径	Exam-ready meaning
differential equation	微分方程	an equation involving a derivative.
rate of change	变化率	a derivative with respect to an independent variable.
constant of proportionality	比例常数	the constant linking proportional quantities.
separable variables	可分离变量	variables that can be placed on opposite sides.
general solution	通解	a family of solutions containing a constant.
particular solution	特解	a solution fixed by an initial condition.
initial condition	初始条件	a known value at a specified input.
equilibrium solution	平衡解	a constant solution with zero rate of change.
growth model	增长模型	a differential equation describing increase.
decay model	衰减模型	a differential equation describing decrease.

4. 单元练习 | Unit practice (共 8 题)

练习 01/08

Question. Form a differential equation: the rate of decrease of M is proportional to M . [2]

练习 02/08

Question. Solve $dy/dx=xy$. [4]

练习 03/08

Question. Solve $dy/dx=2x(y+1)$, given $y=2$ when $x=0$. [5]

练习 04/08

Question. Separate $dy/dx=y(1-y)$. [3]

练习 05/08

Question. Why may division by y lose a solution? [2]

练习 06/08

Question. A model is $P=500e^{(0.04t)}$. Interpret 500 and 0.04. [3]

练习 07/08

Question. Find the equilibrium solutions of $dy/dt=ky(10-y)$. [3]

练习 08/08

Question. Solve $dy/dx=2y$ with $y(0)=5$. [4]

5. 练习答案与解析 | Answers and explanations (共 8 题)

答案 01/08

Correct answer and working. $dM/dt=-kM$, where $k>0$.

答案 02/08

Correct answer and working. For $y \neq 0$, $dy/y=x \, dx$. Hence $\ln|y|=x/2+C$ and $y=Ae^{(x/2)}$. Also $y=0$ is an equilibrium solution.

答案 03/08

Correct answer and working. $dy/(y+1)=2x \, dx$, so $\ln|y+1|=x+C$. Thus $y+1=Ae^x(x)$. Initial condition gives $A=3$, so $y=3e^x(x)-1$.

答案 04/08

Correct answer and working. $dy/[y(1-y)]=dx$. With $1/[y(1-y)]=1/y+1/(1-y)$, integration can proceed.

答案 05/08

Correct answer and working. If $y=0$ is possible, dividing by y excludes it. Test constant values that make the original derivative zero before dividing.

答案 06/08

Correct answer and working. $P(0)=500$ is the initial amount. The positive constant 0.04 is the continuous proportional growth rate per unit time, so $dP/dt=0.04P$.

答案 07/08

Correct answer and working. Set derivative zero: $ky(10-y)=0$, so $y=0$ and $y=10$ are constant equilibrium solutions.

答案 08/08

Correct answer and working. $dy/y=2dx$, so $\ln|y|=2x+C$ and $y=Ae^{(2x)}$. Initial condition gives $A=5$; $y=5e^{(2x)}$.

6. 自学检查清单 | Self-study checklist (共 10 项)

- ☐ 能准确说明 把文字变化率写成方程
- ☐ 能准确说明 分离变量的逻辑
- ☐ 能准确说明 积分常数与初始条件
- ☐ 能准确说明 解释模型而不是只给函数
- ☐ 能准确说明 含 f' / f 或部分分式的分离积分
- ☐ 能准确说明 平衡解与长期行为
- ☐ 能准确说明 从隐式解到显式解
- ☐ 能在没有答案提示时完成本单元全部练习并保留关键步骤
- ☐ 能检查定义域、区间、符号、精度或几何意义，而不是只抄计算器数值
- ☐ 能用英文说明所选公式或方法为何适用于当前条件

3.9 | 复数 | Complex numbers

1. 单元导读 | Unit introduction

本单元严格对应官方 3.9。学习时先建立概念与条件，再推导公式，随后通过完整 worked examples 把方法转化为可复现步骤。P3 的得分关键是：选择正确结构、展示 decisive working、保持 exact values，并检查定义域、区间、参数或几何意义。

与国内课程衔接 | Connection. 国内高中教材已提供相近的代数与微积分基础；Cambridge 更强调多步骤结构题、英文命令词、前后小问的 hence 连接，以及对限制条件的显式说明。

2. 知识点精讲 | Detailed teaching (共 8 个知识点)

知识点 01/08 | Cartesian form 与基本运算 | Cartesian form

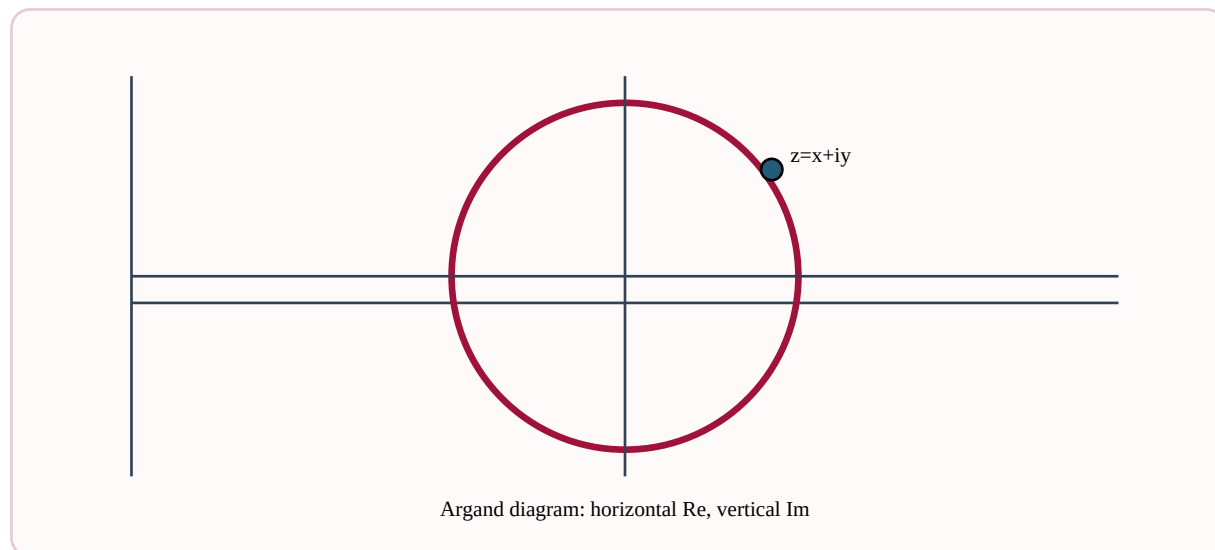
$z=x+iy$ ，其中 $\operatorname{Re} z=x$ 、 $\operatorname{Im} z=y$ 、 $i=-1$ 。两个复数相等当且仅当实部与虚部分别相等。共轭 $\bar{z}=x-iy$ ，且 $z\bar{z}=x+iy=x+y=|z|$ 。

English explanation and method. Worked example. $(3+2i)/(1-i)$ multiply top and bottom by $1+i$: $[(3+2i)(1+i)]/2=(1+5i)/2$. Division must show the conjugate step, not only a calculator display.

知识点 02/08 | 模、辐角与 Argand 图 | Modulus and argument

$|z|=\sqrt{x^2+y^2}$ 是点到原点距离； $\arg z$ 是从正实轴量到向量的角。用 $\tan(y/x)$ 前必须判断象限；主辐角通常取 $-\pi < \theta \leq \pi$ 。

English explanation and method. For $z=-1+\sqrt{3}i$, modulus is 2 and the point lies in quadrant II, so $\arg z=2\pi/3$, not $-\pi/3$. A sketch prevents quadrant errors.



知识点 03/08 | 极形式的乘除法 | Polar form

$z=r(\cos \theta + i \sin \theta)$ 。相乘：模相乘、辐角相加；相除：模相除、辐角相减。结果角度可再调整到题目指定区间。

English explanation and method. Example. $2\operatorname{cis}(\pi/6) \times 3\operatorname{cis}(-\pi/4)=6\operatorname{cis}(-\pi/12)$. Geometrically, multiplication scales lengths by 3 and rotates by $-\pi/4$.

知识点 04/08 | 实系数多项式的共轭根 | Conjugate roots

若实系数多项式有根 $a+bi$ ($b \neq 0$)，则 $a-bi$ 也是根。两者对应实二次因式 $[x-(a+bi)][x-(a-bi)]=(x-a)^2+b^2$ 。

English explanation and method. Example. If $2+i$ is a root, the quadratic factor is $(x-2)+1=x-4x+5$. Divide the polynomial by this real factor to obtain remaining roots.

知识点 05/08 | 复数平方根必须求两个 | Square roots

设 $(a+bi)=u+vi$, 比较 $a-b=u$ 、 $2ab=v$; 再用 $a+b=\sqrt{u+v}$ 辅助。若一个根是 w , 另一个必为 $-w$ 。

English explanation and method. Worked example. For $\sqrt{5+12i}$, solve $a-b=5$, $2ab=12$ and $a+b=13$. Thus $a=9, b=4$ with same signs, giving roots $3+2i$ and $-3-2i$.

知识点 06/08 | 复数轨迹翻译成几何 | Loci on an Argand diagram

$|z-a|=r$ 是以 a 为圆心、半径 r 的圆; $<$ 或 $>$ 给圆内/圆外并处理边界。 $|z-a|=|z-b|$ 是连接 a 、 b 线段的垂直平分线。 $\arg(z-a)=\alpha$ 是从 a 出发、方向角 α 的射线。

English explanation and method. Always label axes Re and Im , the fixed point, boundary type and required region. For an argument locus the starting point a is excluded because $\arg 0$ is undefined.

知识点 07/08 | 复数乘法的几何意义 | Geometric meaning of multiplication

乘以 $re^{i\theta}$ 会把所有点相对原点缩放 r 倍并旋转 θ ; 乘以 i 就是逆时针旋转 $\pi/2$ 。共轭把点关于实轴反射。

English explanation and method. Example. $z \rightarrow (1+i)z$ scales modulus by $\sqrt{2}$ and rotates by $\pi/4$. This explains the result before any Cartesian expansion and provides a check on the computed modulus and argument.

知识点 08/08 | 多个轨迹条件取交集 | Combining loci

若同时给出 $|z-a| \leq r$ 与 $\arg(z-b)$

在某范围, 先分别画圆盘与角区域, 再取共同部分。实线表示包含边界, 虚线表示严格不等式不含边界。

English explanation and method. The point b is excluded from an argument locus because $\arg(z-b)$ is undefined at $z=b$. State this even when the ray appears to start there on the diagram.

3. 核心词汇表 | Core vocabulary (共 12 词)

English term	中文准确口径	Exam-ready meaning
complex number	复数	a number of the form $x+iy$.
real part	实部	the coefficient independent of i .
imaginary part	虚部	the real coefficient of i .
conjugate	共轭	$x-iy$ for $z=x+iy$.
modulus	复数的模	distance from the origin on an Argand diagram.
argument	辐角	directed angle from the positive real axis.
Argand diagram	复平面	a plane with real and imaginary axes.
Cartesian form	代数形式	$x+iy$.
polar form	极形式	$r(\cos \theta + i \sin \theta)$.
conjugate pair	共轭根对	non-real roots $a+bi$ and $a-bi$.
locus	轨迹	the set of points satisfying a condition.
principal argument	主辐角	the selected argument in a standard interval.

4. 单元练习 | Unit practice (共 9 题)

练习 01/09

Question. Express $(3+2i)/(1-i)$ in Cartesian form. [4]

练习 02/09

Question. Find modulus and principal argument of $z=-1+\sqrt{3}i$. [3]

练习 03/09

Question. Multiply $2\text{cis}(\pi/6)$ by $3\text{cis}(-\pi/4)$. [3]

练习 04/09

Question. A real polynomial has root $2+i$. Write the corresponding real quadratic factor. [3]

练习 05/09

Question. Find the square roots of $5+12i$. [5]

练习 06/09

Question. Describe $|z-(2+i)| \leq 3$. [3]

练习 07/09

Question. Describe $|z-1|=|z+3|$. [3]

练习 08/09

Question. Describe geometrically the transformation $z \rightarrow iz$. [2]

练习 09/09

Question. Sketch the region $|z| < 2$ and $\text{Re}(z) \geq 0$. [3]

5. 练习答案与解析 | Answers and explanations (共 9 题)

答案 01/09

Correct answer and working. Multiply by conjugate $1+i$: $(3+2i)(1+i)/2=(1+5i)/2=1/2+(5/2)i$.

答案 02/09

Correct answer and working. $|z|=2$. The point is in quadrant II with reference angle $\pi/3$, so $\arg z=2\pi/3$.

答案 03/09

Correct answer and working. Multiply moduli and add arguments: $6\text{cis}(\pi/6-\pi/4)=6\text{cis}(-\pi/12)$.

答案 04/09

Correct answer and working. The conjugate $2-i$ is also a root. Product $[x-(2+i)][x-(2-i)]=(x-2)+1=x-4x+5$.

答案 05/09

Correct answer and working. Let $(a+bi)=5+12i$. Then $a-b=5, 2ab=12, a+b=13$. Thus $a=3, b=2$ or both negative. Roots: $3+2i$ and $-3-2i$.

答案 06/09

Correct answer and working. The closed disc with centre $2+i$, i.e. point $(2,1)$, and radius 3 on an Argand diagram. The boundary circle is included.

答案 07/09

Correct answer and working. Points equidistant from 1 and -3 on the real axis form the perpendicular bisector of the segment: the vertical line $\operatorname{Re}(z)=-1$.

答案 08/09

Correct answer and working. Multiplication by i preserves modulus and adds $\pi/2$ to the argument: rotation 90° anticlockwise about the origin.

答案 09/09

Correct answer and working. The right half of the open disc centred at the origin with radius 2; the semicircular boundary is excluded, while the imaginary-axis diameter is included.

6. 自学检查清单 | Self-study checklist (共 11 项)

- ☐ 能准确说明 Cartesian form 与基本运算
- ☐ 能准确说明 模、辐角与 Argand 图
- ☐ 能准确说明 极形式的乘除法
- ☐ 能准确说明 实系数多项式的共轭根
- ☐ 能准确说明 复数平方根必须求两个
- ☐ 能准确说明 复数轨迹翻译成几何
- ☐ 能准确说明 复数乘法的几何意义
- ☐ 能准确说明 多个轨迹条件取交集
- ☐ 能在没有答案提示时完成本单元全部练习并保留关键步骤
- ☐ 能检查定义域、区间、符号、精度或几何意义，而不是只抄计算器数值
- ☐ 能用英文说明所选公式或方法为何适用于当前条件

附录 | Appendix

全册共 92 个单元词条、73 道完整练习及同号答案。每个公式均在相关知识点内说明符号、来源或适用条件。

官方范围说明 | Official scope

依据 Cambridge International AS & A Level Mathematics 9709 syllabus for examination in 2028, 2029 and 2030 的 Pure Mathematics 3 (3.1–3.9) 编写。Cambridge International 未参与或背书本资料。